

A Survey on Information Retrieval Systems' Modeling using Term Dependencies and Term Weighting

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Abstract— Information retrieval is a major area of research in various areas. Efficient information retrieval is based on efficient modeling. These models are classified according to two different mathematical modeling approaches: models for term dependency and term weighting. Some models assume term independence, while others do consider this dependency relation. On the other hand, term weighting is a method that tries to index the document in an effective way. The aim of this survey paper is to discuss and esteem different information retrieval models which employ both term dependence/independence, as well as term weighting.

Keywords—Information retrieval system, Mathematical modeling formatting, Term dependency modeling, Term weighting

I. INTRODUCTION

An information retrieval system is a system that able to store, retrieve and maintain information, [18]. Information has many types such as text (include numeric and date data), audio, video and other multimedia. Information retrieval modeling is very important to help researchers in designing and implementing an actual efficient information system. Mathematical modeling can be used in several domains such as geographical areas, medical areas...etc. The model of information retrieval helps the user to predict and explain what a user will find relevant given query. Mathematical retrieval modeling is classified as older models (Boolean and vector space models), probabilistic models (BM-25 and language models) and combining evidence models (inference network and language to rank models). In Fig.1, the classification of mathematical models in IR is shown.

A. Classical Models

This group contains two models: Boolean and region models. In [22], these models provide exact matching. These models use logic operators (and, or, not). These operators are known as "intersection, union, difference". In Fig.2, the different Boolean model operations are shown.

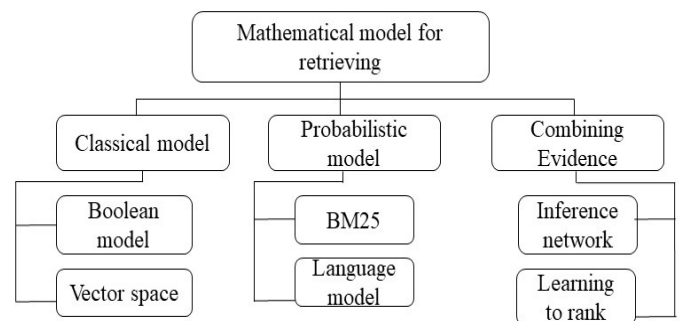


Fig.1. Mathematical modeling classification

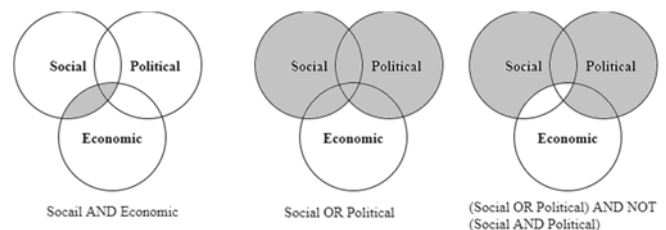


Fig.2. Boolean operation using Venn diagram [22].

Their advantages are given exact match but no ranking retrieved the document. There is a difference between them which is that region model is designed to search for semi-structured data.

The vector space model is used to calculate the similarity "distance" between document and queries through "Euclidean, Manhattan, and cosine similarity. "If cosine angle is zero", then the vector is orthogonal else the degree is zero. The positioning of the query in vector space model is to calculate the centroid of the relevant and irrelevant documents. Moving query towards centroid point of relevant rather than irrelevant meaning it is improving retrieval performance. Both of term weighting's intuition and term's independence are considered the most common disadvantages.

B. Probabilistic Models

Probabilistic models use conditional probability and require two conditions, [22], relevant document and long queries. The relevant document is obtained through computing the probability that contains document terms. Long queries are used to distinguish between term's presence and term's absence in documents. The good choice is that both relevant and non-relevant documents are available. There is a difference between probabilistic approach and knowledge based. The probabilistic approach does not require supplementary data for the data properties and distribution while knowledge based requires some. The result of probabilistic approach is a probability while the result of knowledge based is mostly deterministic.

Two Poisson Model is developing a set of statistical rules to identify indexing term using a mixture of two Poisson, [7, 22]. This model assumes that document was created randomly as a stream of term occurrences. A number of occurrences term frequency (TF) of terms in documents can be modeled by a mixture of two Poisson distribution as follows:

$$p(x = tf) = \lambda \frac{e^{-\mu_1 - (\mu_1)^{tf}}}{tf!} + (1 + \lambda) \frac{e^{-\mu_2 - (\mu_2)^{tf}}}{tf!} \quad (1)$$

Where, x is random variable for the occurrence, λ Proportion of document and μ_1, μ_2 Mean number of occurrence of term.

Their advantages are that they don't need an additional term weighting algorithm to be implemented. It is still one of the best performing term weighting algorithms.

Bayesian network models are a Probabilistic model that use a cyclic directed graph. A directed graph is acyclic if there is no directed path from $A \rightarrow Z$. The presentation of a probability distribution as a directed graph makes it possible to analyze complex conditional independence assumption by graph theory.

The language model has been used in speech recognition, [22]. Speech recognition systems combine two probabilistic models such as acoustic model and language model. The acoustic model might produce instance for decreasing probability order. For example: "good morning", "mood morning" and so on. The language model that determines phrase is much more probable such as "good morning". It occurs more frequently in English rather than other phrases. It builds for each document. The language model, [22], assign a high probability to the word "retrieval". This is indicating that it is a good for retrieval if the query contains this word. It is helpful in that situation which requires models of similarity language or document priors.

Google page rank is a probabilistic model. It is used to determine the quality of pages, [17, 22]. The goal of this algorithm is to track some difficulties with the content-based ranking algorithms of early search engines which use text documents for webpages to retrieve the information with no explicit link relationship between them. It is called as a static ranking function. It is used in the situation that needs modeling of more or less static relations between documents. The advantages of PageRank are that it is a global measure and a query independent as well while the disadvantages are that it is

very efficient to raise your own PageRank and is 'buying' a link on a page with high PageRank as well.

C. Combining Evidence

Learning to rank algorithm is a part of large document retrieval. It is supervised learning task. Data consists of queries and documents. It is the first trained on the training set (represent as feature vectors). It is divided into three approaches (pointwise, pairwise and listwise).

Pointwise: treat ranking is as regular classification. The output of it is a class. The goal of it is to minimize the number of the wrong classification.

Pairwise: It transforms ranking into pointwise classification. The goal of it is to decrease the number of pairs which are ranked out of order.

Listwise: It is similar to pairwise. It reduces loss function.

Finally, there are two measures which are usually used to assess the effectiveness of retrieval method. The first one is called precision rate which is equal to retrieved relevant document that is actually retrieved. Secondly, recall rate. It is equal to retrieved relevant documents that are actually successfully relevant. If we want to raise precision, then we have to narrow queries. If we want to increase the recall, then we broaden the query. So the relation between precision and recall is an inverse relation. F-measure is the third one that combines precision and recall.

The rest of this paper is organized as follows: modeling of information retrieval systems in section II and conclusion in section III.

II. MODELING OF INFORMATION RETRIEVAL SYSTEM'S

The main two problems in vector space model are assuming terms are independent and term weighting is intuitive but informal as shown in Fig.3.

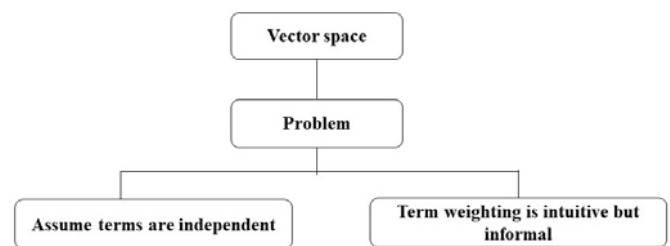


Fig.3. Problems of vector space

A. Term Dependency Modeling

There are many models that assume terms are independent such as Binary independence language, ontology..etc. Now, dependency modeling is discussed. In Fig.4, the dependency modeling is shown.

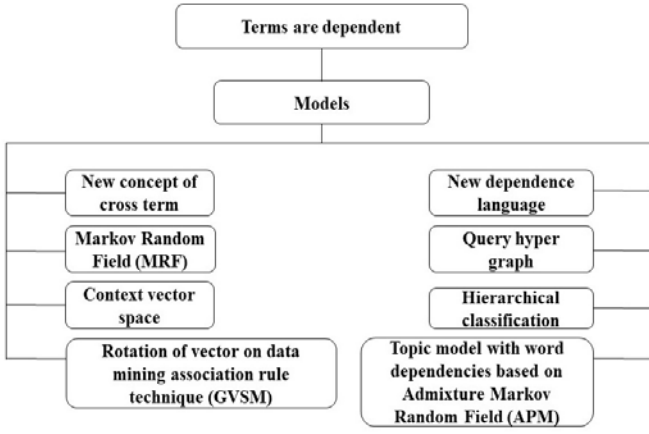


Fig.4. Dependency modeling

In [1], a new concept of the cross term with bigram, n-gram and kernel functions were proposed. Firstly, Cross Term with bigram was used as Basis, Then n-gram was generalized for n queries such that ($n \geq 3$). Shape functions are used to describe the impact of the query term. The functions should be satisfied proprieties (non-negative, continues, symmetric, monotonic and identified). Shape function has seven kernel functions that satisfy effect of query term such as (Gaussian kernel, triangle function, circle function, cosine kernel, Quadratic kernel, Epanechnikov kernel and triweight kernel). The Gaussian kernel was widely used especially in statistics and machine learning. Triangle, circle, and cosine were used in genomic graphics. These functions were used to find the distance. If the distance is raised then the effect is weak. All of these models were applied on (CRTER2).

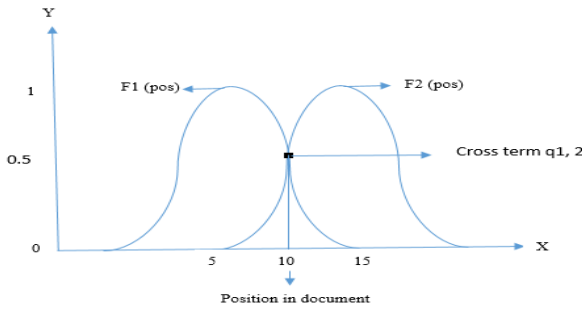


Fig.5. An example of bigram cross term.

The shapes of several equations are following:

Gaussian kernel:

$$kernel(u) = \exp\left| \frac{-u^2}{2\sigma^2} \right|, \quad (2)$$

Triangle kernel:

$$kernel(u) = \left(1 - \frac{u}{\sigma}\right) \cdot 1_{\{u \leq \sigma\}}, \quad (3)$$

Circle kernel:

$$kernel(u) = \sqrt{1 - \left(\frac{u}{\sigma}\right)^2} \cdot 1_{\{u \leq \sigma\}}, \quad (4)$$

Cosine kernel:

$$kernel(u) = \frac{1}{2} \left[1 + \cos\left(\frac{u\pi}{\sigma}\right) \right] \cdot 1_{\{u \leq \sigma\}}, \quad (5)$$

Quadratic kernel:

$$kernel(u) = \left(1 - \left(\frac{u}{\sigma}\right)^2\right) \cdot 1_{\{u \leq \sigma\}}, \quad (6)$$

Epanechnikov kernel:

$$kernel(u) = \left(1 - \left(\frac{u}{\sigma}\right)^2\right) \cdot 1_{\{u \leq \sigma\}}, \quad (7)$$

Triweight kernel:

$$kernel(u) = \left(1 - \left(\frac{u}{\sigma}\right)^3\right) \cdot 1_{\{u \leq \sigma\}}, \quad (8)$$

Bag of words, Bi-term and many dependencies were introduced in, [2]. Some bi-term models exist such as (Bag of Words, Markov random field, Divergence from Randomness and BM5 Model) and many term dependency model such as (BM25-Span Model, Positional language model (PLM)). All of them assume that terms are dependent. BM25-Span model evaluates each span by distorting between width and number of the query term in each span. PLM determines occurrence if each query in the document to neighbor locations. Then kernel functions are used to determine the frequency of each term in Document. The kernel that used is Gaussian kernel functions.

In [3], a new dependence language extends to language based on unigram was used. Most dependencies do not lead to a development in effectiveness in retrieving large things. There are two reasons for this:-

Firstly, the difficulty to estimate dependencies in large scale. Secondly, the integration of both single words and dependencies in weighting schema. The bi-gram language model is better than unigram model. New model expresses term dependencies as a cyclic, planar and an undirected graph. The query is generated from documents in two stages:-

Linkage generated is a term generated. This study is applied on different six collection dataset on different models such as binary independence retrieval (BIR), unigram (UG), dependence model (DM), bi-gram language model(BG) and bi-term language model (BT1 and BT2). These comparisons lead to improvement in precision.

In [4], Query hyper graphs have also been used to capture complex dependencies between query concepts statements. Queries are acted as vertices and edges. The distance between edges is called “dependencies”. Vertex is corresponding to query. Query hyper graph is derived a ranking function that treats with concept and dependency concept. The model is proposed in this research integrates three main characteristics:

1. Model arbitrary term dependencies as a concept.
2. It uses a passage level evidence to model dependencies between these concepts.
3. It assigns a weight to both concepts: concept dependencies and weight to important features.

Experiments use newswire and web corpora collection. This framework improves the effectiveness of several retrieval models.

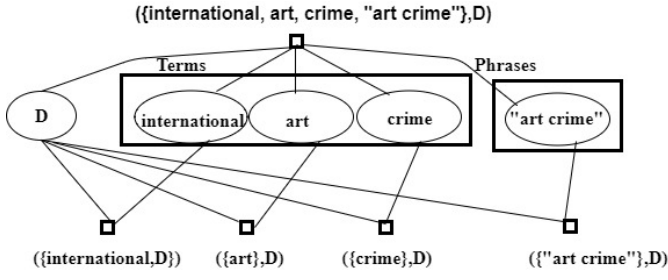


Fig.6. An example of a hyper graph representation for the query “international art crime”, [4].

Markov Random Field “MRF” has been used in [5, 6] with a new variation in, [21] to scout full independencies, sequential dependencies, and full dependencies. In [5], MRF is constructed from the undirected graph. Nodes represent random variable and edges define independence between variables. Steps of MRF are to Construct graph for query term dependencies and define a set of potential function and rank document.

Full independencies (FI): Terms which occur and don’t affect by other terms.

Sequential dependencies (SD): terms are dependencies between neighbors.

Full dependencies (FD): Terms which depend on each other and show a complete graph and capture longer range dependencies.

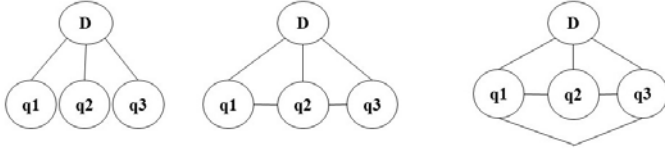


Fig.7. An example of MRF for three query term dependencies (left “FI”, middle “SD”, right “FD”) [5].

The potential function is very important to complete efficiency. The good potential function is assigned a high value to most compatible with each other under given distribution our result shows that modeling dependencies improve effectiveness across a range of TREC collection. Sequential dependencies using ordered features are more effective on the smaller collection while the full Ones are the best for larger

$$\log \phi_t(c) = \lambda_t \log p(q_i | D) \\ = \lambda_t \log [(1 - \alpha_D) \frac{tf_{q_i, D}}{|D|} + \alpha_D \frac{cf_{q_i}}{|C|}] \quad (9)$$

Where $|D|$ is the total number of terms in documents, $|C|$ is the length of collection, $tf_{(q_i, D)}$ is the number of times occur in document, cf_{q_i} is the number of times occur in entire collection and α is the smoothing parameter.

In [21], a new variation of Markov Random Field that relies on BM-25 was proposed. MRF is one of dependency models. It takes the most attention in recent years. It gives a clear effectiveness not performance because of high computational

costs. The new model is applied on TREC8, GOV2, Clueweb09-Category-B collections. It reduces costs by up 60% and keeps the effectiveness as the same with no loss.

In [7], a new topic model based on an admixture of Poisson Markov Random Field (APM) was introduced. APM model dependencies between words are opposed to previous independent topic model such as PLSA. Poisson MRFs (PMRFs) provides JOINT distribution over multivariate count data. The model PMRF (θ, Θ) is defined as follows:

$$pr_{PMRF}(x | \theta, \Theta) \propto \exp\{\theta^T x + x^T \Theta x - \sum_{s=1}^p \ln(x_s!)\} \quad (10)$$

Based on the dependency of parameter Θ , if Θ is negative then the dependency is rarely co-occurred. Else, often co-occur. The experiments are applied on Grolier encyclopedias that provide visually appealing and interpretable Results.

In [8], the model is an extension of the vector space model using the association rule of data mining techniques to discover set of terms that co-occur in the document collection. The advantages of vector space model are partial matching, good ranking, simple, fast and it is the most used definition of term weight. In this research, incorporate information of correlation among terms in the collection of vector space model improves effectiveness. Generalized vector space model (GVSM) is another extension of vector space model. In GVSM, terms can be non-orthogonal and represented by a small component called “minterms” with binary weight. In VSM evaluates the degree of similarity between query and documents based on distance. The similarity between two vectors is calculated as follows:

$$sim(d_j, q) = \frac{d_j \cdot q}{|d_j| \times |q|} = \frac{\sum_{i=1}^t w_{i,j} \cdot w_{i,q}}{\sqrt{\sum_{i=1}^t w_{i,j}^2} \sqrt{\sum_{i=1}^t w_{i,q}^2}} \quad (11)$$

And the confidence equation is:

$$confidence(A \Rightarrow B) = p(B/A) = \frac{freq(A \cap B)}{freq(A)} \quad (12)$$

Apriori for association rules represents a frequent pattern of data. This algorithm identifies co-occurrence among terms. Vector space with association rule is the confidence of terms. Confidence chooses to compute the new angle between term vectors. Term vectors are brought close together according to rules. If 90 then orthogonal angle between vectors K_i , K_j then rotation occurs only in K_i , K_j not modified. The angle is given by:

$$\theta_{ij} = 90(1 - c_{ij}) \quad (13)$$

Where θ_{ij} is a new vector between (K_i, K_j) and c_{ij} is a confidence of association rule $K_i \rightarrow K_j$

The experiments were made with four reference collection named CACM, CYSTIC FRIBOSIS (CFC), CISI and third text retrieval conference (TREC-3). The result shows the effectiveness of retrieval. The result with association rules is better rather than VSM and improves precision leading up to 31%.

In [9], a novel method of question retrieval “hierarchical question classification” was used. People can ask the question with natural language instead of segment words and answer to question posted online or offline. Question retrieval is very

important which attracts a great deal of interest from the research and community. Overview of question retrieval is illustrated in the Fig.8.

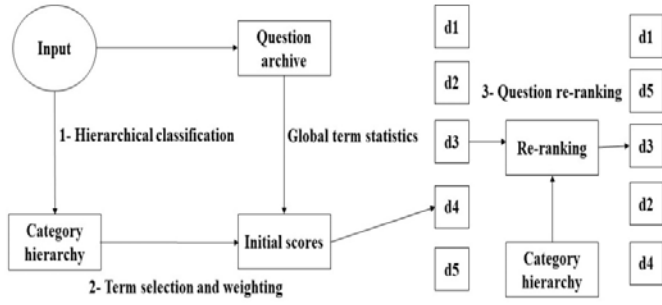


Fig.8. Question retrieval workflow.

Experimental result on Yahoo dataset shows the effectiveness of this method compared to another state of art method.

Term context vector model was used in [10], based on co-occurrence of term in the same document. Vectors are used to calculate context vector for a document. A context vector is obtained from words occurring close to an entity in a text. Vectors represent the context of single occurrence. Set of term context vector can be represented by nxn matrix as follows:

$$\begin{pmatrix} c_{11} & \dots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

The use of term context vectors allows two different interpretation of index 1- syntactic interpretation, 2- semantic description. Once a term correlation matrix has been generated, an initial document into context vector is released. The following equation making is as shown below:

$$\vec{d} = \frac{\sum_{j=1}^n w_{ij} \frac{\bar{t}_j}{|t_j|}}{\sum_{j=1}^n w_{ij}} \quad (14)$$

B. Term Weighting Methods

Term weight is another factor that effects on the result. There are different methods to measure weighting. These are classified as supervised (TFIDF) and unsupervised (Gain Ratio and Confidence weight) that is shown in Fig.9.

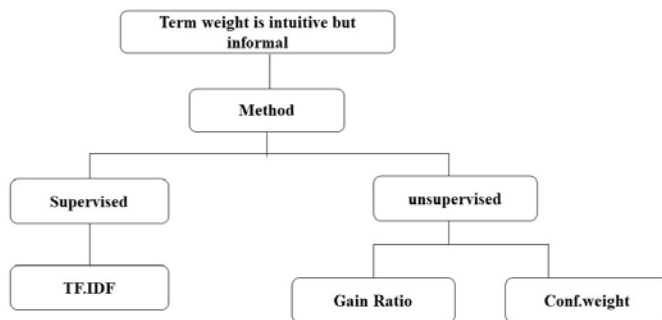


Fig.9. Term weighting methods

It can improve retrieval performance. Many different terms weighting schemes have been proposed in, [10, 11, and 12]. In

[12], term frequency (TF) and inverse document frequency (IDF) are the most used and most effective weighting scheme. Term frequency is based on an occurrence of the term in a document. Inverse document frequency is based on a term which occurs in many documents. The TF*IDF is calculated as follows:

$$idf(t_i) = \log \frac{N}{n_i} \quad (15)$$

$$w_i = tf_i * \log \frac{D}{df_i} \quad (16)$$

Where D is the number of documents, t_i is the term count and df_i is the number of documents containing term (i).

It is applying on the standard dataset and found that TF*IDF is high relevance. Deviation of term weighting is discussed in, [10]. Terms that occur in all documents with a different frequency are more important than terms occur in all documents once. A Kind of deviation assesses that different occurrence should improve retrieval performance. The variation of TF*IDF standard across document vector shows as follows:

The modified average mean deviation of term context vector:

$$tcvmamd(t_i) = 1 + \frac{\sum_{j=1}^n |\frac{c_{ij}}{\text{mean } t_j} - 1|}{n} \quad (17)$$

The modified variance of term context vectors:

$$tcvmamd(t_i) = 1 + \frac{\sum_{j=1}^n (\frac{c_{ij}}{\text{mean } t_j})^2}{n-1} \quad (18)$$

The combination of idf and tcvmamd:

$$tcvmamd(t_i) = 1 + idf(t_i) \frac{\sum_{j=1}^n |\frac{c_{ij}}{\text{mean } t_j} - 1|}{n} \quad (19)$$

The experiments are applied to four different collections (MED, CRANFIELD, CISI, and CACM).

A variation of standard TF*IDF can be found in, [13]. Term weighting is split into three categories: local, global and normalization. Local weight is calculated according to a number of occurrence terms in document or query. Global weight is a number of occurrence terms in the entire collection. Normalization is done after local and global weight. It is not necessary because it doesn't affect on ranked document list. List of establishing local weight formulas is given in TABLE 1, TABLE 2, and TABLE 3:

TABLE 1. LOCAL WEIGHT FORMULA [13]

Formula	Name	Abbr
1 if $f_{ij} > 0$ 0 if $f_{ij} = 0$	Binary	BNRY
f_{ij}	Within document frequency	FREQ
$1 + \log f_{ij} > 0$ if $f_{ij} > 0$ 0 if $f_{ij} = 0$	log	LOGA

$\frac{1 + \log f_{ij}}{1 + \log a_{ij}}$	$if f_{ij} > 0$	Normalized log	LOGA
0	$if f_{ij} = 0$		
$0.5 + 0.5 \left(\frac{f_{ij}}{x_{ij}} \right)$	$if f_{ij} > 0$	Augmented normalized term frequency	ATF1
0	$if f_{ij} = 0$		

TABLE 2. GLOBAL WEIGHT FORMULA [13]

Formula	Name	Abbr
$\log \left(\frac{N}{n_i} \right)$	Inverse document frequency	IDFB
$\log \left(\frac{N - n_i}{n_i} \right)$	Probabilistic inverse	IDFP
$1 + \sum_{j=1}^N \frac{f_{ij} \log \frac{f_{ij}}{F_i}}{\log N}$	Entropy	ENPY
$\frac{F_i}{n_i}$	Global frequency IDF	IGFF
1	No global weight	NONE

TABLE 3. NORMALIZATION FACTORS [13]

Formula	Name	Abbr
$\frac{1}{\sqrt{\sum_{i=0}^m (G_i L_{ij})^2}}$	Cosine normalization	COSN
$\frac{1}{(1 - slope) + slope l_j}$	Pivoted unique normalization	PUQN
1	None	NONE

The term weighting is based on question retrieval model shows the relationship between terms pairs when calculating their weight. To overcome this problem, novel term weighting scheme by incorporate dependency relation between two pairs is proposed in, [14]. First, construct dependency graph and compute relation strength between each term. Second, refine initial term weight. The undirected graph ensures that every term pair has a dependency relation path. If the path is shorter, then it reflects stronger relation shown in the Fig.10.

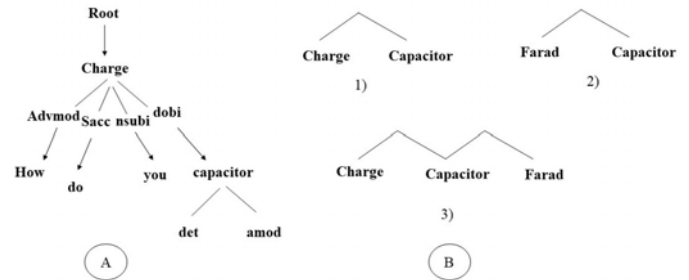


Fig.10. a) Dependency parsing tree and b) Dependency relation path [14].

This method is applied using large real-world data set from Yahoo and the result is to improve retrieval effectiveness.

A new method of term weighting is introduced in, [15]. This method is called confidence weight. It is based on the statistical estimation of the importance of the word. It also has benefit making feature selection. Confidence weight is used as an alternative to TF*IDF. It has the ability to perform well if no features are conducted. If the feature is irrelevant, then weight it gets from confidence weight is low. Three data sets are used to test such as Reuters that made of categories related to the business news report, Ohsumed that comes from large text collections and rarely used with all available categories and documents of term weight are shown in TABLE 4.

TABLE 4. COMPARISON BETWEEN METHODS

Method	Stable	Sensitive	Accuracy
Gain information ratio	More	Less	Less
Confidence weight	More	More	More
TF*IDF	Less	More	Less

In [16], a new term weighting method was proposed. This method doesn't use information of query but uses similarity information between documents. To map similarity of the cluster into weight, we use information gain ratio (IGR). If amount of information of word in cluster increases after the cluster is partitioned into sub-clusters then word used to determine the structure of sub-clusters. The cluster consists of two sub-clusters. One sub-cluster is the cluster of retrieved documents which will be partitioned into smaller clusters. Another is the rest of database. Structured Similarity describes relation among documents. The clustering retrieval document is based on IGR are shown as follows in Fig.11.

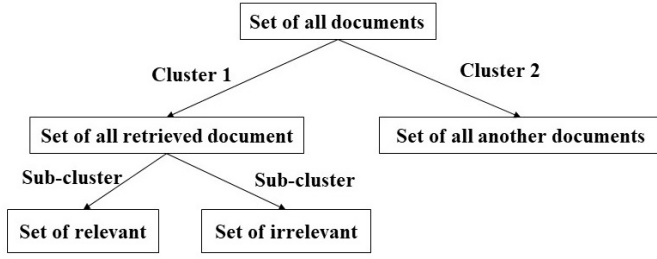


Fig.11. Clustering retrieval documents.

There are three types of term weighting (TF, IDF, and IGR). TF*IDF depends on the distribution of each word in documents. IGR depends on the structure of document cluster and to analyze similarity among retrieved through maximum distance algorithm. The distance is calculated as Euclidean distance as follows:

$$d(D_i, D_j) = \sqrt{\sum_k (weight_{ik} - weight_{jk})^2} \quad (20)$$

$$weight_{ik} = tf(D_i, w_k)idf(w_k) \quad (21)$$

$$tf(D_i, w_k) = \frac{freq(D_i, w_k)}{|D_i|} \quad (22)$$

$$idf(w_k) = \log_2 \frac{N}{df(w_k)} \quad (23)$$

Where N is the total number of documents, $|D_i|$ is the number of morphemes in D_i , $freq(D_i, w_k)$ is the frequency of the word w_k in D_i and $df(w_k)$ is the document frequency of the word.

IGR is the measure used in decision tree learning algorithm C4.5 to choose the best attribute. The dataset used is TSC. It has 12 topics. Each topic has one query and 50 retrieved. After that improving accuracy, time and effectiveness were found.

Clustering is unsupervised learning technique. It helps to group data into classes. The cluster is a collection of objects that similar within the same cluster and dissimilar in another cluster. One of the most popular techniques is k-mean algorithm. It is found in [17]. Document clustering is one of the fastest growing research. It is unsupervised learning technique that helps to organize similar document in classes to improve retrieval. Before applying this algorithm, there is preprocessing in [18] should be done such as (tokenization, preprocessing and feature extraction). This algorithm is applied on 20_new group dataset. In [19], Attributed k-mean method task is to map distribution attribute in the applied on 20_new group dataset. In [19], Attributed k-mean method task is to map distribution attribute in the dataset. There are three measures used to evaluate the quality of clustering. First, F-measure: It combines between recall and precision. Secondly, Entropy: it is used to measure the goodness of unnested cluster. It determines how homogenous cluster is. So the relation between homogeneity and entropy is inverse. Higher homogeneity has lower entropy.

$$E(c_i) = -\frac{1}{\log c} \sum_{h=1}^k \frac{n_i^h}{n_i} \log \left(\frac{n_i^h}{n_i} \right) \quad (24)$$

Where c_i is the cluster and n_i is the size.

Thirdly, purity: it evaluates the coherence of cluster. To achieve high quality, maximize F-measure, purity and minimize entropy.

$$p(c_i) = \frac{1}{n_i} \max_h (n_i^h) \quad (25)$$

Traditional TF-IDF considers the weight of term frequency and inverse document frequency not a concern with the weight of other feature of the word. A new method called TF-IDF adaptive position in [20] determines the weight of position of the word. TF-IDF-AP is applied on Chinese expression. F-measure of TF-IDF-AP has improved by 12.9% compared with classical TF-IDF. The formula for the position of the first occurrence described as:

$$first\ position(word) = \frac{FPBeforecount(word)+1}{\sum_{i=0}^n count(word_i) - FPBeforecount(word)} \quad (26)$$

The formula for the position of last occurrence described as:

$$last\ position(word) = \frac{LPAftercount(word)+1}{\sum_{i=0}^n count(word_i) - LPAftercount(word)} \quad (27)$$

The formula for Adaptive weight described as:

$$position\ weight = \frac{1}{First\ position(word) + Last\ position(word)} \quad (28)$$

The formula of weight described as:

$$weight(word, Doc) = \frac{TF*IDF*position\ weight}{\sqrt{\sum_{word, Doc} [TF*IDF*position\ weight]^2}} \quad (29)$$

III. CONCLUSION

By studying various dependencies modeling and term weighting methods, it is attempted to conduct a survey of dependencies models and weighting methods. Unigram, Bigram and n-gram models and new dependence language are based on unigram to improve precision and effectiveness on six different collections of TREC dataset. Bigram cross Term (CRTER2) records improvement on all six standards of TREC. Retrieval using hierarchical classification, hyper graph and association rule of data mining record improving in effectiveness but on different data collection such as yahoo answer data set, newswire and web corpora respectively. In addition to association rules, context vector model improves average precision. Also, Context vector improves performance.

Methods to measure term weighting are TF*IDF and Information Gain Ratio which improves performance, accuracy and time respectively. The new method is called “confidence weight” which is a replacement of TF*IDF. It is more stable, more sensitive, and more accurate rather than the other methods.

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